Type MS · Vigenere cipher

Answer 1 of 1 · Question 3 part c · 4 marks

2021 JUN · Paper 02 · extract 1 of 8 · find part c in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark
3 (b)	B1	Award up to 2 marks for a linked explanation such as: • The number of keys are limited/only one shift used (1) making it easy to use brute force to decrypt (1) • It can be easy to find commonly used letters (e.g. E) (1) and guess the key (1)	Accept alternative similar wording.	(2)

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2021 JUN · Paper 02 · extract 2 of 8 · find part c in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark
3 (c)		Award 1 mark each up to a maximum of 4 for: <table><tr><td>Encrypted letter</td><td>f</td><td>l</td><td>m</td><td>k</td><td>t</td><td>r</td><td>w</td><td>h</td><td>e</td><td>e</td></tr><tr><td>Keyword letter</td><td>t</td><td>h</td><td>i</td><td>r</td><td>t</td><td>y</td><td>t</td><td>h</td><td>i</td><td>r</td></tr><tr><td>Decrypted letter</td><td>m</td><td>e</td><td>e</td><td>t</td><td>a</td><td>t</td><td>d</td><td>a</td><td>w</td><td>n</td></tr></table>		
	C1	Award 1 mark from: • flmktrwhee mapped to thirtythir (1) • Flmktr mapped to thirty (1)		
	C2	At least one letter decrypted correctly (1)		
	C3	At least one word decrypted correctly (1)		
	C4	Decrypted message 'meet at dawn' (1)		(4)

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Question	Mp	Answer	Additional Guidance	Mark
3 (d)(i)	D1	Award 1 mark for: • cipher letter / a single encrypted letter (1)	Do not accept word/message/text	(1)

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		• cipherLetter / a single encrypted letter (1)		(1)
3 (d)(ii)	D2	Award 1 mark for any of: • keywordLetter • plaintextLetter	Ignore case	(1)

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Answer 1 of 1 · Question 3 part c · 4 marks

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		• plaintextletter		(1)
3 (d)(iii)	D2	Award 1 mark for any of: • subprogram that is already defined • subprogram that is already written • subprogram that is already compiled • subprogram that can be called without having to write code for it	Must be clear they are referring to subprogram provided by the language itself	(1)

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Answer 1 of 1 · Question 3 part c · 4 marks

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Type MS · Vigenere cipher

Answer 1 of 1 · Question 3 part c · 4 marks

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Question	MP	Answer	Additional Guidance	Mark
3 (a)	A1	Get plaintext and store in plaintext variable	Accept alternative wording Line numbers may be different compared to the examples shown When testing the completed code use lowercase for the input	(1)
	A2	Get key and store in key variable		(1)
	A3	Validate key (must repeat until a valid key has been input)		(1)
	A4	Open file to write/append		(1)
	A5	Write cipher text		(1)
	A6	Close file		(1)
	A7	Displays ciphertext		(1)
	A8	Executing and producing correct output to file and screen (must have A3)		(1)
Code examples				
Python				

```
15
16 #       Add your code to get the plaintext and convert it to lowercase
17 plaintext = input("Enter the plaintext using lowercase letters: ")
18 plaintext = plaintext.lower()
```

```
# Add your code to get the key and make sure the key is between 1 and 25
while key < 1 or key > 25:
    key = int(input("Enter the key - a number between 1 and 25 "))
```

```
39 # Add your code to write the ciphertext to a text file
40 cipherFile = open("Cipher.txt", "w")
41 cipherFile.write(ciphertext)
42 cipherFile.close()
```

```
40
41 #       Add your code to display the ciphertext
42 print("The ciphertext message is:",ciphertext)
43
```

Type MS · Vigenere cipher

Answer 1 of 1 · Question 3 part c · 4 marks

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